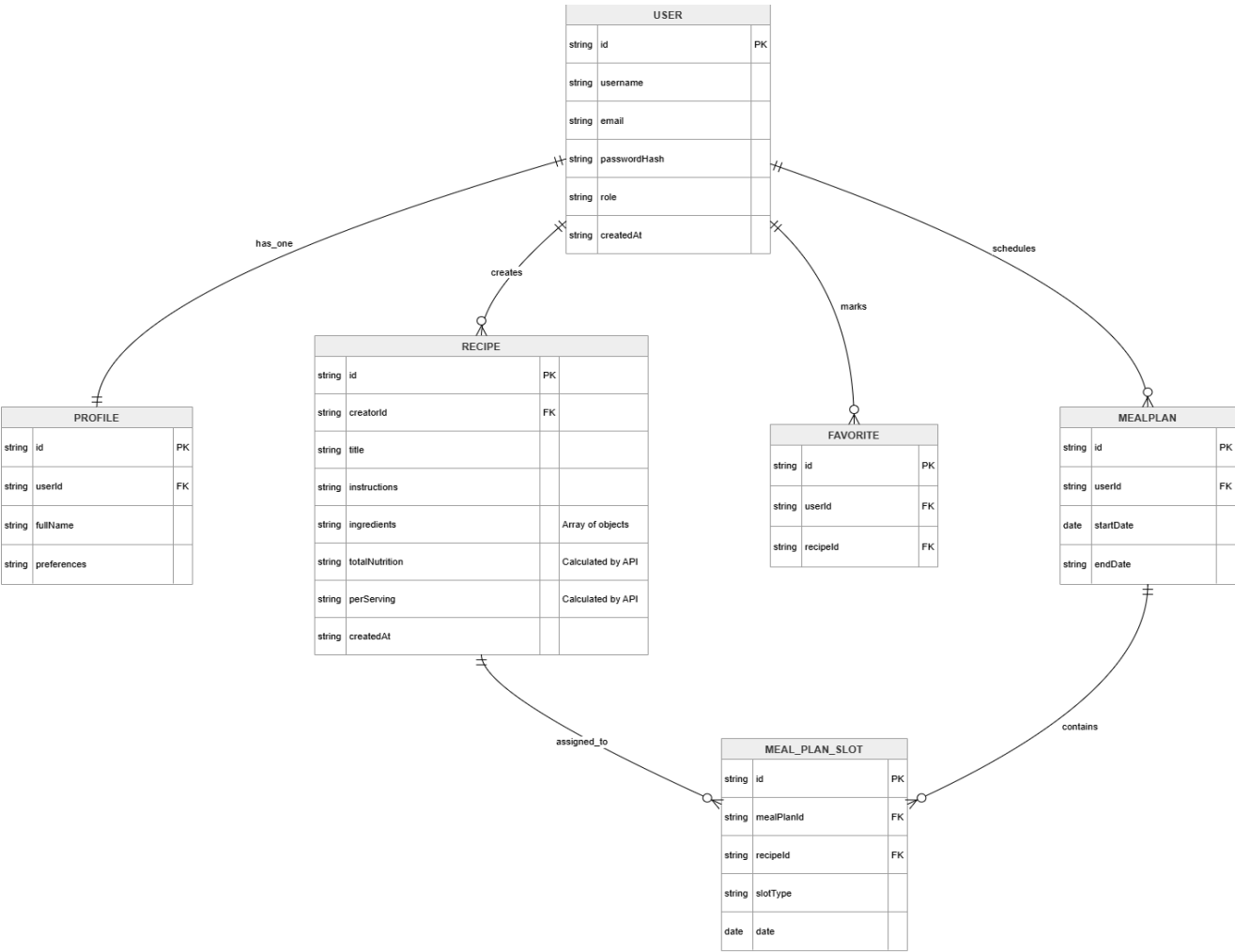


Database Schema

The application uses **MongoDB** (NoSQL document database) managed through the **Mongoose ODM**. Data is stored in the **vitality-kitchen** database across **4 collections**: users, recipes, mealplans, and meals.



Collection: users

Field	Type	Required	Unique	Default	Description
_id	ObjectId	Auto	Yes	Auto	Primary key
name	String	Yes	No	—	User's display name
email	String	Yes	Yes	—	Login email; unique across all users
password	String	Yes	No	—	Bcrypt-hashed password (never plain text)
calorie_goal	Number	No	No	2000	Daily calorie target (kcal)
dietary_prefs	[String]	No	No	[]	Dietary tags e.g. Vegan, Keto
favorites	[ObjectId]	No	No	[]	References to favorited Recipe._id docs
avatar	String	No	No	null	Base64-encoded profile image or URL

Collection: recipes

Field	Type	Required	Description
_id	ObjectId	Auto	Primary key
user	ObjectId	Yes	Reference to User._id who created the recipe
isUserCreated	Boolean	No (def: true)	false for system-seeded library recipes
title	String	Yes	Recipe name
description	String	No	Short description
cuisine	String	No	Cuisine type (e.g., Italian, Asian)
category	String	No	Enum: Breakfast, Lunch, Dinner, Snack
Field	Type	Required	Description

dietaryTags	[String]	No	Dietary labels (e.g., Vegan, Gluten-Free)
servings	Number	Yes	Number of servings the recipe yields
prepTime / cookTime	Number	No	Preparation / cooking time in minutes
image	String	No	Base64-encoded image string
ingredients	[Object]	No	Array of ingredient sub-documents
steps	[String]	No	Ordered cooking instructions
totalNutrition	Object	No	{calories, protein, carbs, fat} — sum of all ingredients
perServing	Object	No	totalNutrition / servings
ratings	[Object]	No	Array of {user: ObjectId, rating: 1–5} sub-documents
averageRating	Number	No (def: 0)	Computed average of all ratings
createdAt / updatedAt	Date	Auto	Mongoose timestamps

Collection: mealplans

Field	Type	Required	Description
_id	ObjectId	Auto	Primary key
user	ObjectId	Yes	Reference to User._id
startDate	Date	Yes	UTC-midnight Monday of the planned week
week	Object	No	7 day keys, each with Breakfast/Lunch/Dinner slots (Recipe refs or null)
createdAt / updatedAt	Date	Auto	Mongoose timestamps

Compound Unique Index: (user, startDate) — prevents duplicate weekly plans for the same user. Each slot in week.{day}.{slot} holds a Reference to Recipe._id or null when empty. All 21 meal slots are populated on read.

Week Object Structure

```
week: { monday: { Breakfast: ObjectId|null, Lunch: ObjectId|null, Dinner: ObjectId|null },
  tuesday: { ... }, wednesday: { ... }, thursday: { ... }, friday: {
  ... }, saturday: { ... }, sunday: { ... } }
```

Relationship Summary

Relationship	Type	Mechanism
User → Recipes	One-to-Many	Recipe.user — ObjectId reference
User → MealPlans	One-to-Many	MealPlan.user — ObjectId reference
User ↔ Favorites	Many-to-Many	User.favorites[] — Array of ObjectId refs to Recipe
MealPlan ↔ Recipes	Many-to-Many	MealPlan.week.*.* — ObjectId refs in 21 week slots
Recipe ↔ Raters	Many-to-Many	Recipe.ratings[].user — Embedded sub-document